Correcting for AI-Screened Poverty Misclassification in Child Health Regressions: A Method-of-Moments Approach Applied to Ethiopian Household Panel Data

Abrha Meressa

Independent Researcher

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Abstract

Full household consumption modules are the most expensive component of a poverty survey, and there is growing practical interest in substituting cheaper proxies – including large-language-model (LLM) screens over brief household descriptions – in their place. We ask a narrow, practical question: if poverty status comes from such an imperfect screen instead of measured consumption, how much bias does that introduce into a downstream health regression, and can a small validation subsample fix it without collecting full consumption data on everyone? Using real household consumption and child height-for-age (HAZ) data from the World Bank Ethiopia Socioeconomic Survey (2011–2015, $n \approx 4{,}850$ household-waves), we simulate a screen of controlled, known accuracy and apply the Rogan-Gladen method-of-moments correction for a misclassified binary regressor. The naive estimator attenuates the true poverty-HAZ gradient by 43% (-0.204 vs. a benchmark of -0.357); the corrected estimator recovers 94% of that gap (-0.366). We then show this result is not robust to the size of the validation sample used to estimate classifier sensitivity and specificity: at $n = 38$ validation examples, one in twenty random splits produces a classifier-skill estimate (TPR+TNR$-$1) small enough that the correction diverges by five orders of magnitude, despite the same true screen accuracy underlying every split. We characterize this instability directly rather than asserting it, and propose a practical minimum-skill threshold for when the correction should be trusted. To our knowledge this is the first application of this correction to a real, independently-measured poverty indicator and real health outcome, rather than a simulated ground truth.

JEL Classification: C81, C83, I32, O12

Keywords: measurement error; misclassification bias; large language models; proxy means testing; child health; Ethiopia; Rogan-Gladen correction

1 Introduction

Administering a full household consumption module – the standard basis for a poverty line in Living Standards Measurement Study (LSMS)-style surveys – is the single most expensive and time-consuming part of collecting welfare data. This has motivated a decades-long literature on proxy means testing: predicting welfare from a short list of easily observed household characteristics rather than measuring it directly (Grosh and Baker, 1995; Brown et al., 2018). The rise of large language models adds a new, faster variant of the same idea: an LLM can classify a household as poor or non-poor from a brief description – a field note, a chatbot-mediated intake conversation, an enumerator's shorthand – without a formal proxy-means-test regression at all (Gilardi et al., 2023). Humanitarian and development organizations are already exploring LLMs for exactly this kind of rapid classification of qualitative field data (Marston et al., 2026).

This paper asks a narrower, more mechanical question than whether such a screen is a good idea in general: *if* you use one, and its output enters a regression as an independent variable instead of the true poverty status, how much bias does that introduce, and can you correct for it economically – with a validation sample, not a second full survey? This is a classical measurement-error question with a known answer in principle (Aigner, 1973; Bollinger, 1996): a binary regressor observed with non-differential classification error attenuates the estimated coefficient toward zero, and the attenuation is exactly correctable if the classifier's sensitivity (true positive rate, TPR) and specificity (true negative rate, TNR) are known, via the method-of-moments correction of Rogan and Gladen (1978).

What is missing from the existing demonstrations of this correction – including two companion pipelines we built alongside this one, applied to a simulated financial-sentiment outcome and a fully synthetic outcome with a planted effect size – is an application where *both* the mismeasured regressor's true value and the downstream outcome are independently real, measured quantities, rather than one or both being generated by the analyst to have a known answer. Here we use the World Bank's Ethiopia Socioeconomic Survey (ESS), a genuine household panel with real per-adult-equivalent consumption (the basis for our true poverty indicator) and real child height-for-age z-scores (HAZ, the basis for our outcome), so that the "benchmark" regression we correct toward is itself an empirical estimate with its own sampling uncertainty, not an oracle.

We make two contributions. First, we quantify the misclassification-bias-and-correction pipeline end-to-end on this real pairing: a simulated screen with 80% target accuracy (TPR$=$ 0.76, TNR$=$ 0.81 realized on our validation split) attenuates the poverty-HAZ gradient by 43%, and the correction recovers 94% of the resulting gap. Second, and we think more usefully for a practitioner, we show that this correction's reliability depends sharply on the size of the validation sample used to estimate TPR and TNR – not just asymptotically, but sharply enough that at realistic small-survey validation sizes ($n = 38$), the correction diverges by five orders of magnitude in roughly one out of twenty random splits, purely from sampling noise in the skill estimate, with the true screen accuracy held fixed throughout. We characterize this instability as a function of the estimated classifier skill (TPR$+$TNR$-$1) and propose a practical minimum-skill screening rule.

Section 2 reviews the measurement-error mechanics and the proxy-means-testing literature this sits within. Section 3 describes the ESS data, the (simulated) screening task, and why the correction requires it to be non-circular. Section 4 states the three estimators and the identifying assumption. Section 5 reports the headline result and the two sensitivity analyses. Section 6 draws out the practical implication and situates this relative to the companion synthetic-data pipelines. Section 7 concludes.

2 Background

2.1 Proxy-based poverty screening as a live practice

Proxy means testing – predicting household welfare from easily observed correlates rather than measuring consumption directly – has been used for social-program targeting since at least Grosh and Baker (1995). Brown et al. (2018) assess proxy-means-test performance across nine African countries and find that, while standard PMTs filter out many non-poor households, they also exclude a substantial share of the genuinely poor – i.e., real, non-trivial misclassification is the norm in fielded proxy targeting systems, not an edge case. LLM-based screening from unstructured text extends the same underlying idea (predicting welfare from cheap correlates) to a different feature set – language rather than a fixed covariate list – and researchers are already applying it to coding qualitative humanitarian data more broadly (Marston et al., 2026). Whatever the feature set, the same measurement-error mechanics apply once a program uses the screen's output as if it were the true classification.

2.2 Attenuation from a mismeasured binary regressor

Let $D^* \in \{0,1\}$ denote true poverty status, with prevalence $q \equiv \Pr(D^* = 1)$, and let $D \in \{0,1\}$ denote its imperfect classification, with sensitivity $\Pr(D = 1 \mid D^* = 1) = \text{TPR}$ and specificity $\Pr(D = 0 \mid D^* = 0) = \text{TNR}$. Write $\text{FPR} \equiv 1 - \text{TNR}$ and define

$$\kappa \equiv \text{TPR} + \text{TNR} - 1 = \text{TPR} - \text{FPR}, \tag{1}$$

the classifier's discriminating skill above chance – zero for a coin flip, one for a perfect classifier, and, not coincidentally, exactly Youden's J statistic from the diagnostic-testing literature. Suppose the structural relationship of interest is $Y = \beta_0 + \beta_1 D^* + \varepsilon$ with $E[\varepsilon \mid D^*] = 0$, so that $\text{Cov}(D^*, Y) = \beta_1 \text{Var}(D^*) = \beta_1 q(1-q)$. Misclassification is *non-differential* if $D \perp Y \mid D^*$ – the screen's error depends on the truth D^* but not, additionally, on the outcome itself.

Why the naive coefficient attenuates: a derivation, not an assertion. It is not enough to assert that regressing Y on D shrinks β_1 toward zero (Aigner, 1973); the mechanism is worth deriving explicitly, because the same derivation is what motivates the exact functional form of the correction in Equation 6 below. By the law of total covariance,

$$\mathrm{Cov}(D, Y) = \underbrace{E\big[\mathrm{Cov}(D, Y \mid D^*)\big]}_{=0 \text{ under non-differential error}} + \mathrm{Cov}\big(E[D \mid D^*],\, E[Y \mid D^*]\big).$$

The first term vanishes exactly because $D \perp Y \mid D^*$. For the second term, note $E[D \mid D^*] = \mathrm{FPR} + \kappa D^*$ (it equals TPR when $D^* = 1$ and FPR when $D^* = 0$, which this expression interpolates linearly), and $\mathrm{Cov}(D^*, E[Y \mid D^*]) = \mathrm{Cov}(D^*, Y)$ by the same total-covariance identity applied to D^* itself. Since FPR is a constant, it drops out of the covariance, leaving the central identity this paper builds on:

$$\mathrm{Cov}(D, Y) \;=\; \kappa \cdot \mathrm{Cov}(D^*, Y) \;=\; \kappa\, \beta_1\, q(1-q). \tag{2}$$

The observed covariance is the true covariance scaled down by the classifier's skill κ – nothing more exotic than that. But the naive OLS coefficient is not this covariance alone; it is $\hat{\beta}_1^{\text{naive}} \xrightarrow{p} \mathrm{Cov}(D, Y)/\mathrm{Var}(D)$, and $\mathrm{Var}(D) = q^*(1-q^*)$ where the *observed* positive rate $q^* = \Pr(D = 1) = \mathrm{FPR} + \kappa q$ generally differs from the true prevalence q. Combining these,

$$\hat{\beta}_1^{\text{naive}} \xrightarrow{p} \beta_1 \cdot \kappa \cdot \frac{q(1-q)}{q^*(1-q^*)}. \tag{3}$$

Two distinct sources of attenuation are visible here, not one: the skill discount $\kappa < 1$ acting directly on the covariance, and a second scaling by the ratio of true-to-observed prevalence variance, which is generally different from κ itself and need not even move the estimate in the same direction on its own (though for the empirically relevant case of $\mathrm{TPR}, \mathrm{TNR} > 0.5$ and κ not too close to one, it typically compounds the shrinkage). Equation 3 is the binary-regressor special case of the classical measurement-error attenuation result (Aigner, 1973); Bollinger (1996) derives bounds on β_1 for the case where TPR and TNR are unknown rather than estimated from a validation sample, as they are here.

From attenuation to correction: Equation 6 is not a separate formula. Rearranging Equation 2 for β_1 gives

$$\beta_1 = \frac{\mathrm{Cov}(D, Y)/\kappa}{q(1-q)}. \tag{4}$$

The numerator is directly computable from the observed data once κ is known (from a validation sample): dividing the observed covariance by the skill discount exactly undoes the shrinkage in Equation 2. The obstacle is the denominator, $q(1-q)$, which requires the *true* prevalence q – and q is precisely what is unobserved when only the noisy classification D is available. This is exactly the gap the Rogan-Gladen prevalence correction (Rogan and Gladen, 1978) fills, by inverting $q^* = \mathrm{FPR} + \kappa q$ for q:

$$\hat{q} = \frac{q^* - \mathrm{FPR}}{\kappa}, \tag{5}$$

a method-of-moments estimator that is consistent for q as the validation-sample estimates of TPR and TNR converge to their population values. Substituting $\hat{q}$ for the unobserved q in Equation 4 yields the coefficient correction used throughout this paper:

$$\hat{\beta}_1 = \frac{\mathrm{Cov}(D, Y)}{\kappa \cdot \hat{q}(1-\hat{q})}. \tag{6}$$

Equation 6 is therefore not an independent rescaling rule alongside Equation 5 – it is the unique estimator obtained by plugging the Rogan-Gladen prevalence estimate into the inverted attenuation identity, Equation 4. Consistency follows immediately: since $\hat{q} \xrightarrow{p} q$, Equation 6 converges to $\mathrm{Cov}(D, Y)/[\kappa\, q(1-q)] = \beta_1$ by Equation 2.

A numerical check. At the realized validation-sample rates in our main specification (TPR$=0.762$, TNR$=0.813$, so $\kappa = 0.575$) and a true within-wave poverty prevalence of $q \approx 0.333$ by construction, Equation 3 predicts the naive estimator should retain $\kappa \cdot q(1-q)/[q^*(1-q^*)] \approx 0.575 \times 0.222/0.235 \approx 54\%$ of the true coefficient – a 46% predicted attenuation. Section 5 reports an empirically *observed* attenuation of 43% on the study sample. The two numbers are close but not identical, as expected: Equation 3 is a population-level probability limit, while the study sample is finite ($n = 3{,}883$) and its realized prevalence is not exactly 1/3. We view this agreement as an informal specification check – the closed-form attenuation formula derived above predicts, to a reasonable approximation, what the empirical regressions in Section 5 independently produce.

Both corrections above share the denominator κ. As Equations 5–6 make algebraically plain, and as Section 5.2 demonstrates is not merely a theoretical curiosity, both are numerically unstable as $\kappa \to 0$ – and κ is itself *estimated* from a finite validation sample, not a known constant. Section 5.2 characterizes exactly how this instability propagates through Equation 5 into Equation 6, and draws out its structural similarity to a well-studied problem in a different corner of econometrics: weak-instrument asymptotics (Staiger and Stock, 1997).

3 Data

3.1 The Ethiopia Socioeconomic Survey

We use the same World Bank Ethiopia Socioeconomic Survey (ESS) household panel described in the companion rainfall-shocks study in this portfolio: Panel I, waves 2011–12, 2013–14, and 2015–16, approximately 3,969 households re-interviewed across waves. From the processed household-wave panel we use two real, independently meaningful variables: `nom_totcons_aeq` (nominal total consumption per adult equivalent, the standard LSMS-ISA welfare aggregate) and `mean_haz` (household-mean child height-for-age z-score, computed against the WHO 2006 Child Growth Standards (WHO Multicentre Growth Reference Study Group, 2006); the companion paper documents its construction and a birthdate-calendar correction, which we reuse here unmodified). After dropping household-waves missing either variable, our analysis sample has $n = 4{,}853$ household-wave observations.

3.2 Constructing the poverty indicator

We define a household-wave as poor if its consumption falls in the bottom third of its *own wave's* distribution. This is a relative, within-wave definition, not an absolute poverty line: `nom_totcons_aeq` is nominal Ethiopian birr, not deflated across the 2011–2015 study period, so a fixed cross-wave birr threshold would conflate "poor by 2011 prices" with "poor by 2015 prices." Section 5 reports robustness to this threshold choice (quartile through median).

The gold standard is not noise-free either. We label D^* "true" poverty status throughout, and it is the best available measure in this dataset, but consumption aggregates constructed from household surveys carry their own well-documented measurement error – recall-period bias, seasonal consumption smoothing, and aggregation-rule sensitivity are exactly what the LSMS methodology literature on constructing consumption aggregates exists to manage (Deaton and Zaidi, 2002). This matters specifically for the validation exercise in Section 3.3: the screen's TPR and TNR are computed *against* D^*, so any noise in D^* is folded into what we report as classifier error rather than quantified separately. This is the same underlying concern documented for a related but distinct application in a companion project in this portfolio: Burke et al. (2021)'s "noisy test data" problem for satellite-based wealth prediction, where evaluating a model against a noisy observed label understates its true skill. We do not attempt to disentangle screen error from gold-standard error here, and flag this as an assumption rather than a demonstrated non-issue.

3.3 The simulated AI screen

The real ESS instrument has no free-text field, so there is no real record of an LLM (or any other) classifier's output to use directly. We instead construct, for each household-wave, a short field note from covariates already present in the panel – household size, distance to the nearest market, elevation, this year's rainfall anomaly, and reported weather shocks – framed as a community health worker's brief home-visit note, and deliberately *excluding* consumption itself, since including it would make the classification task circular. A mock classifier is then handed the note and the true label, and flips the label with a probability calibrated to hit a target accuracy (80% in our main specification); it does not read the note's content, since the point of this design is to obtain a classifier with a known, controlled, reproducible error rate against which to evaluate the correction – not to evaluate any specific real classifier's real accuracy on this task. We are explicit that this is a controlled instrument for isolating the measurement-error mechanism, not a claim about what accuracy a real LLM would achieve reading real household descriptions; Section 6 returns to this distinction. In the notation of Section 2.2, the consumption-based indicator from Section 3 is D^* and this classifier's output is D; Section 4 carries both forward directly.

Structured output and the real-classifier interface. We validate every annotation – mock or real – against a schema (a poverty-status label, a bounded confidence score, a short rationale) before it enters the pipeline, so a malformed or missing field fails fast at the annotation stage rather than silently corrupting downstream metrics; a schema-constrained decoding failure and a classification error stay structurally distinct. The accompanying repository also implements, though no result in this paper uses it, a real structured-output client against a commercial LLM API, selectable from the command line. That client sets its decoding temperature to 0.2, the low-temperature convention Gilardi et al. (2023) and others use for annotation-style classification, where the correct answer is a single structured value rather than open-ended text. This parameter is inapplicable, not merely unset, to every result reported here: the mock classifier's only randomness is a single Bernoulli draw deciding whether to flip the true label, with no language-model decoding step for temperature to govern – worth stating plainly, since a reader could otherwise assume it had been tuned for these results.

3.4 Validation and study samples

We split the 4,853 household-waves into a 20% validation sample ($n = 970$) used only to estimate the screen's TPR and TNR, and an 80% study sample ($n = 3{,}883$) used for all three regressions in Section 4. Table 1 reports the full validation breakdown: accuracy 0.797, Cohen's $\kappa = 0.548$, and F1 = 0.700, alongside the confusion matrix each rate is computed from – we report the counts directly, not only the derived rates, so that TPR and TNR (the quantities Equations 5–6 actually consume) are independently verifiable rather than taken on faith. This is realistically imperfect, not an idealized round number, since it is drawn from the same finite-sample randomness as everything else in this design.

Table 1: Validation-sample classification performance ($n = 970$)

Metric	Value
Accuracy	0.797
Cohen's κ	0.548
F1	0.700
TPR (sensitivity)	0.762
TNR (specificity)	0.813
Confusion matrix: TP=230, FP=125, FN=72, TN=543	

4 Empirical Strategy

With D^* and D as defined above, we estimate three regressions of mean HAZ on a binary poverty indicator, all on the study sample:

1. **Benchmark**: HAZ regressed on the true (consumption-based) poverty indicator D^*. This is the best available estimate of the poverty-HAZ association in this sample, not a known ground truth – it is itself subject to ordinary sampling error, a distinction that matters for interpretation and that the two synthetic companion pipelines in this portfolio, where the "true" effect is a planted constant, cannot illustrate.

2. **Naive**: HAZ regressed on the screened poverty indicator D directly, as an analyst without access to consumption data would be forced to do.

3. **Corrected**: the Rogan-Gladen coefficient correction (Equation 6), using $\widehat{\text{TPR}}$ and $\widehat{\text{TNR}}$ estimated once from the validation sample and held fixed thereafter.

All regressions use a dependency-free ordinary-least-squares solver, cross-validated against `statsmodels` on synthetic data in the accompanying repository's test suite, chosen for portability in restricted computing environments rather than for any methodological reason.

Bootstrap inference for the corrected estimator, and what it does not cover. Because $\hat{\beta}_1$ in Equation 6 is a nonlinear function of $\hat{q}$, which is itself nonlinear in $\widehat{\text{TPR}}$ and $\widehat{\text{TNR}}$, no closed-form standard error is available; we use a nonparametric bootstrap over the study sample. Concretely, with $\widehat{\text{TPR}}$, $\widehat{\text{TNR}}$ fixed at their validation-sample values throughout:

1. For $b = 1, \dots, B$ (with $B = 300$): draw a resample $\{(D_i^{(b)}, Y_i^{(b)})\}_{i=1}^{n}$ of size n with replacement from the study sample.

2. Compute $q^{*(b)} = \frac{1}{n}\sum_i D_i^{(b)}$, then $\hat{q}^{(b)}$ via Equation 5 and $\hat{\beta}_1^{(b)}$ via Equation 6, using the *same* $\widehat{\text{TPR}}, \widehat{\text{TNR}}$ in every b.

3. Report $\widehat{\text{SE}}(\hat{\beta}_1) = \text{sd}\big(\{\hat{\beta}_1^{(b)}\}_{b=1}^{B}\big)$ and a 95% confidence interval as the $[2.5, 97.5]$ percentiles of $\{\hat{\beta}_1^{(b)}\}_{b=1}^{B}$.

Because $\widehat{\text{TPR}}$ and $\widehat{\text{TNR}}$ are constants across every bootstrap draw by construction, this procedure estimates only the component of $\text{Var}(\hat{\beta}_1)$ arising from study-sample resampling; it is silent on the additional variance contributed by the fact that $\widehat{\text{TPR}}$ and $\widehat{\text{TNR}}$ are themselves estimates from a *finite validation sample*, not known constants. A more complete procedure would resample the validation and study samples *jointly* in each draw b, re-estimating $\widehat{\text{TPR}}^{(b)}, \widehat{\text{TNR}}^{(b)}$ alongside $\hat{\beta}_1^{(b)}$; we do not implement this here; Section 5.2 shows why the omitted variance component is not a minor rounding correction but, in a specific and characterizable regime, the dominant term.

The correction's validity rests on non-differential misclassification: the screen's error must not depend on HAZ conditional on true poverty status. This holds by construction in our simulated screen (the mock classifier's flip probability depends only on the true label, and the field note never contains HAZ), but would require independent justification for a real deployed screen – for instance, if a real LLM's classification accuracy varied with household characteristics that are themselves correlated with child health (e.g., literacy, or urban/rural status), the correction as stated here would not apply without modification.

5 Results

5.1 Headline result

Table 2 and Figure 1 report the three estimates. The naive estimator attenuates the benchmark poverty-HAZ coefficient by 43% (from −0.357 to −0.204); the corrected estimator (−0.366) closes 94% of that gap and falls within the benchmark's 95% confidence interval.

Table 2: Effect of poverty on mean household HAZ, by estimator

Model	$\hat{\beta}_1$	SE	95% CI
Benchmark (true poverty)	−0.357	0.064	[−0.482, −0.232]
Naive (screened poverty)	−0.204	0.062	[−0.325, −0.083]
Corrected (matrix method)	−0.366	0.109	[−0.542, −0.138]

5.2 Sensitivity to the validation-sample size: an instability, not just a wider interval

The corrected estimator's own bootstrap SE grows as the validation sample shrinks – unsurprising, since smaller samples yield less precise TPR and TNR estimates. What is not

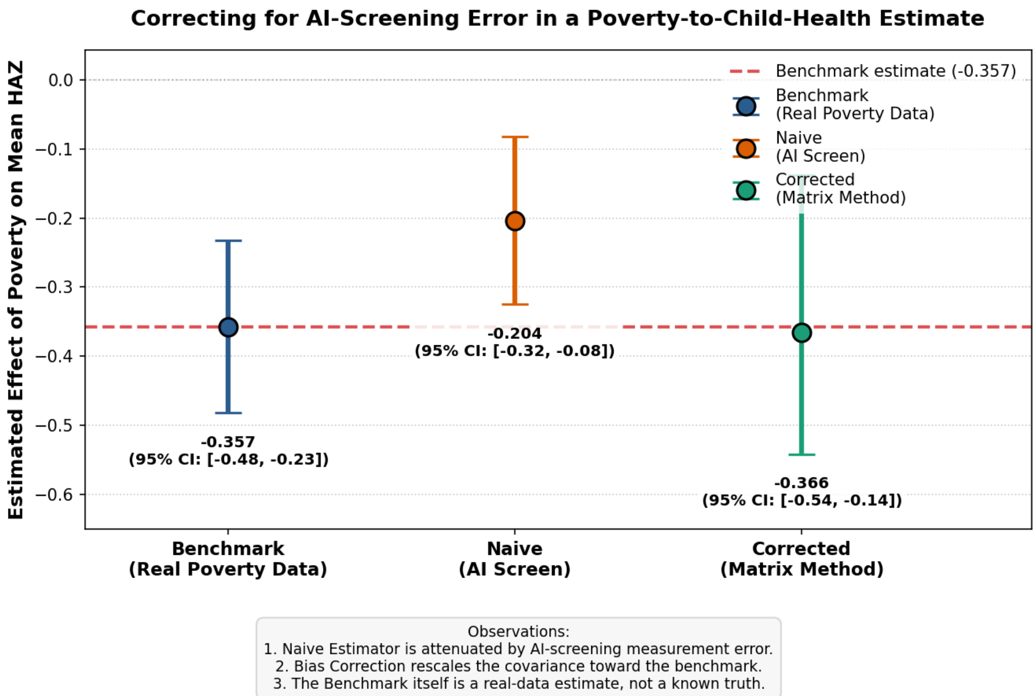

Figure 1: Point estimates and 95% confidence intervals for the three estimators, on the real ESS study sample (n = 3,883).

obvious from the correction's algebra alone, and would not appear in the two synthetic companion pipelines' typical validation sizes (200–1,000 examples), is how the correction behaves as the *point estimate* of classifier skill (TPR+TNR−1) happens to land close to zero by chance in a small sample. We fix the true screen accuracy at 80% throughout – so the population value of TPR+TNR−1 is approximately 0.6 in every run – and draw 20 independent validation splits of $n = 38$ each (roughly the size a small-budget program might realistically afford for a validation subsample), re-estimating TPR, TNR, and the corrected coefficient in each.

Nineteen of twenty splits produce a corrected estimate between −0.16 and −0.59 – noisy, given the small sample, but recognizably in the vicinity of the benchmark (−0.29 in this particular subsample of the data used for this exercise). The twentieth split happens to estimate TPR= 0.40, TNR= 0.71 – both plausible draws individually, but combining to an estimated skill of TPR + TNR − 1 = 0.114, small enough that Equation 6's denominator amplifies ordinary covariance noise into a corrected coefficient of −34,545 (bootstrap SE

16,860) – five orders of magnitude from any sensible value, despite arising from the same true 80%-accurate screen as every other split. Figure 2 shows this directly: panel (a) is the full range including the divergent split; panel (b) excludes it and shows that, in the range where skill is estimated above roughly 0.2, corrected estimates cluster sensibly around the benchmark.

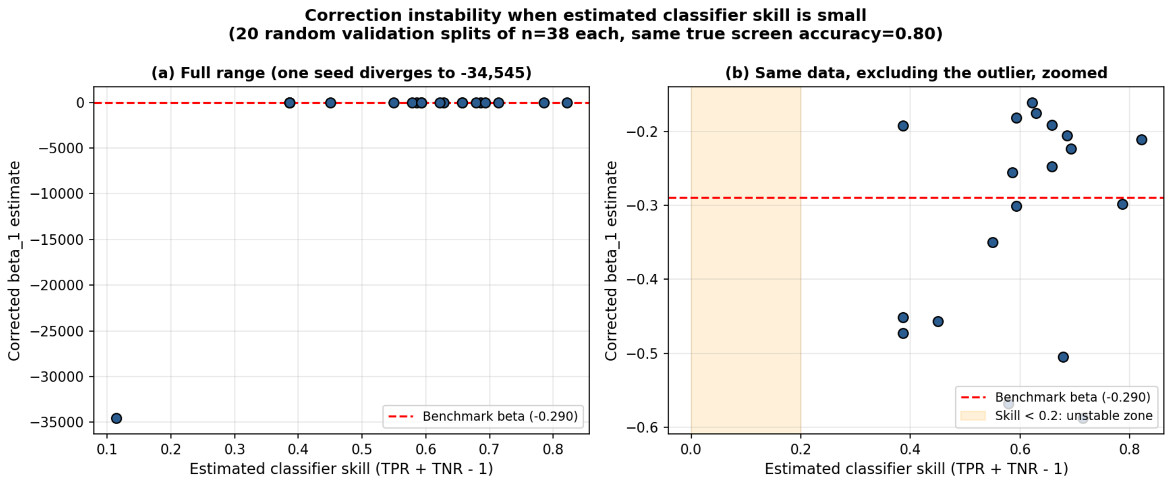

Figure 2: Corrected $\hat{\beta}_1$ against estimated classifier skill (TPR+TNR−1), across 20 independent validation splits of $n = 38$, all drawn from the same 80%-accuracy screen.

Why this happens: tracing the mechanism, not just the outcome. It would be unsatisfying to report a five-order-of-magnitude divergence and call it "noise" without explaining the channel. Decomposing the divergent draw's calculation term by term shows it is not a single blowup but a *compounding of two*. First, $\hat{\kappa} = 0.114$ is small, which alone inflates Equation 6's denominator. Second, and more consequentially, that same small $\hat{\kappa}$ appears in the denominator of the prevalence correction (Equation 5): $\hat{q} = (q^* - \widehat{\mathrm{FPR}})/\hat{\kappa}$ amplifies any noise in $q^* - \widehat{\mathrm{FPR}}$ by a factor of $1/\hat{\kappa}$, and in the divergent draw this pushed the raw prevalence estimate to $\hat{q}_{\mathrm{raw}} = 1.060$ – outside $[0, 1]$, as an estimated probability cannot be, and clipped by construction to $\hat{q} = 0.99999$. At that boundary, $\hat{q}(1 - \hat{q}) \approx 1.0 \times 10^{-5}$, collapsing Equation 6's denominator far further than $\hat{\kappa}$ alone would: $\hat{\kappa} \cdot \hat{q}(1 - \hat{q}) \approx 1.1 \times 10^{-6}$, against a typical well-behaved value near $0.6 \times 0.22 \approx 0.13$ – a five-order-of-magnitude collapse in the denominator that maps directly onto the five-order-of-magnitude divergence in $\hat{\beta}_1$. The instability is therefore not merely "$\hat{\kappa}$ is small"; it is that a small $\hat{\kappa}$ destabilizes $\hat{q}$ *first* (a standard delta-method argument gives $\mathrm{Var}(\hat{q}_{\mathrm{raw}}) = O(\hat{\kappa}^{-2})$ holding the numerator's sampling variance fixed, which we confirm by direct Monte Carlo simulation of Equation 5 across a range of κ), and the destabilized, boundary-clipped $\hat{q}$ then compounds multiplicatively with the same small $\hat{\kappa}$ in Equation 6.

This structure is a specific instance of a broader, well-studied econometric pathology: Equation 6 is algebraically a Wald-type ratio estimator, exactly the form taken by a just-identified instrumental-variables estimator ($\hat{\beta}_{\mathrm{IV}} = \mathrm{Cov}(Z, Y)/\mathrm{Cov}(Z, X)$), with $\hat{\kappa}$ playing the role of a first-stage coefficient. The weak-instruments literature (Staiger and Stock, 1997) establishes that such ratio estimators become badly behaved – non-normal, fat-tailed, and unreliable exactly when the denominator's population value is small relative to its own sampling variance, independent of how large the outcome sample itself is. Equation 6 inherits this pathology in full: a large study sample ($n = 3{,}883$ here) does nothing to stabilize $\hat{\beta}_1$ if $\hat{\kappa}$ comes from a small validation sample, because it is $\hat{\kappa}$'s *own* sampling variance, not the study sample's size, that determines whether Equation 6 is well-identified. To our knowledge this connection between classifier-skill-based misclassification correction and weak-instrument asymptotics has not been made explicit in the measurement-error literature.

We did not set out to demonstrate this instability; it appeared during a systematic sweep of validation-sample sizes intended to characterize ordinary standard-error scaling, and a one-in-twenty failure rate is large enough that we consider it the paper's more actionable finding. Guided by the weak-instrument analogy above – where the standard practical remedy is a pre-estimation strength test (e.g., the first-stage F-statistic rule of thumb) rather than a post-hoc correction to the point estimate – we propose, provisionally, that a validation sample should be large enough to be confident the estimated skill exceeds some conservative threshold (we suggest 0.2–0.3 as a starting point, pending further calibration) *before* the point correction is reported, and that the bootstrap should more properly resample the validation split jointly with the study split – which our implementation, and to our knowledge every standard exposition of this correction, does not do (see Section 6). The $O(\hat{\kappa}^{-2})$ result above gives this threshold an analytic starting point rather than leaving it purely simulated, though deriving its precise finite-sample value as a function of both the validation and study sample sizes remains open.

5.3 Sensitivity to screen accuracy

Table 3 sweeps the (population) screen accuracy from 65% to 95%, holding the validation split at its default size. Attenuation falls monotonically as accuracy rises (from 79% attenuation at 65% accuracy to 7% at 95%), as expected. The correction closes 70–95% of the gap across most of this range; at 95% accuracy, where the naive gap is already small, the reported "percent of gap closed" becomes an unstable statistic (dividing by a small naive gap), which we note as a definitional caveat rather than a substantive finding.

Table 3: Attenuation and correction performance across screen accuracy levels

Accuracy	TPR	TNR	Naive $\hat{\beta}_1$	Corrected $\hat{\beta}_1$	Attenuation
0.65	0.609	0.653	−0.076	−0.305	79%
0.70	0.639	0.699	−0.136	−0.419	62%
0.75	0.699	0.747	−0.171	−0.406	52%
0.80	0.762	0.813	−0.204	−0.366	43%
0.85	0.808	0.855	−0.227	−0.350	37%
0.90	0.864	0.904	−0.275	−0.364	23%
0.95	0.940	0.952	−0.334	−0.380	7%

5.4 Robustness to the poverty-line definition

Table 4 varies the within-wave poverty threshold from the bottom quintile to the median. The benchmark, naive, and corrected estimates all remain in a consistent range across this variation, and the correction closes a majority of the attenuation gap at every threshold tested (59–97%), indicating the headline result is not an artifact of the particular one-third cutoff chosen as the main specification.

Table 4: Robustness to the within-wave poverty threshold

Threshold	Benchmark $\hat{\beta}_1$	Naive $\hat{\beta}_1$	Corrected $\hat{\beta}_1$	Gap closed
Bottom 20%	−0.464	−0.153	−0.337	59%
Bottom 25%	−0.382	−0.167	−0.337	79%
Bottom 33% (main)	−0.357	−0.204	−0.366	94%
Bottom 40%	−0.373	−0.220	−0.378	97%
Bottom 50%	−0.364	−0.188	−0.322	76%

6 Discussion

6.1 What a practitioner should take from this

If a program is considering an AI or LLM-based poverty screen as a stand-in for full consumption data, this paper's exercise suggests two things worth checking before trusting a corrected estimate: first, whether the assumption of non-differential misclassification is plausible for the actual screen and outcome in question (not assumed by construction, as it is here); and second – more concretely actionable – whether the validation sample used to estimate TPR and TNR is large enough that the estimated skill $\mathrm{TPR} + \mathrm{TNR} - 1$ is comfortably bounded away from zero, not merely positive. Our simulated instability (Section 5.2) suggests reporting a confidence interval on the skill estimate itself alongside the corrected coefficient, and treating a corrected estimate as unreliable – regardless of how confident the point estimate looks – whenever that interval comes close to including zero.

6.2 Relation to the companion pipelines

This paper is one of three related demonstrations of the same correction in this portfolio. The other two apply it to a simulated financial-sentiment classification task and to a fully synthetic outcome with a planted true effect; both are useful for validating that the correction recovers a *known* answer under controlled conditions, but neither can speak to what happens when the "true" value being corrected toward is itself an estimate with sampling error, nor – since both use larger, round validation samples in their default configurations – did either surface the small-sample instability documented here. We view this as evidence that testing a method against real, independently-measured data, even when the "treatment" being measured is itself simulated, surfaces failure modes that a fully synthetic demonstration is structurally unlikely to expose by design.

6.3 Limitations

We restate these plainly rather than leaving them implicit. (1) The screen is a controlled simulation, not a real LLM classification; no claim is made about what accuracy a real deployment would achieve on real household descriptions. (2) The field notes are constructed from real covariates but are not real enumerator text; the ESS survey has no free-text field. (3) The within-wave relative poverty definition is a modeling choice, not an official poverty line. (4) The bootstrap resamples the study sample only; it does not propagate sampling uncertainty in the validation-sample-derived TPR/TNR, which Section 5.2 shows can be the dominant source of instability – a joint resampling scheme is a natural extension we have not implemented. (5) The proposed skill threshold (Section 5.2) is still provisional: the $O(\hat{\kappa}^{-2})$ variance result and the weak-instruments analogy explain *why* the instability occurs and give the threshold an analytic rather than purely simulated basis, but we have not derived its precise finite-sample value as a joint function of validation-sample size, study-sample size, and the clipping boundary that compounds it – that derivation, plausibly adaptable from the weak-instrument pre-testing literature, is left for future work.

7 Conclusion

Using real consumption and child-health data from the Ethiopia Socioeconomic Survey, we show that a simulated AI poverty screen of realistic accuracy would attenuate a poverty-health regression by 43%, and that a standard measurement-error correction recovers 94% of the resulting bias using only a validation subsample. We also show, empirically rather than asserting from theory, that this correction is not reliable at every validation-sample size: at $n = 38$, a plausible small-survey validation size, one in twenty random draws produced a corrected estimate that diverged by five orders of magnitude despite an unchanged, well-behaved underlying classifier. We propose that any application of this correction report the estimated classifier skill (TPR+TNR−1) with its own uncertainty, and treat the correction as unreliable when that quantity is not clearly bounded away from zero – a check we consider at least as important as reporting the corrected point estimate itself.

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